

Miruna Oprescu

📍 New York, NY ✉ miruna@cs.stonybrook.edu 🌐 mirunaoprescu.com in [antonia-miruna-oprescu](https://antonia-miruna-oprescu.github.io)

Research Interests

Themes: Trustworthy AI, causal inference, learning and decision-making under uncertainty.

Methods: Statistical machine learning, reinforcement learning, sequential learning and inference, robust estimation and uncertainty quantification.

Applications: Healthcare, climate and Earth systems, neuroscience, energy and industrial systems.

Academic Appointments

Stony Brook University

Sep 2026 – Present

SUNY PRODiG+ Fellow, Department of Computer Science

- Building TILDE Lab: **T**rustworthy **I**nference, **L**earning, **D**ecision-making, and **E**valuation.
- Planned tenure-track transition to Assistant Professor in 2028 following successful review.

Education

Cornell University (Cornell Tech)

2021 – 2026

Ph.D. in Computer Science. Advisor: Nathan Kallus

M.S. in Computer Science (2024, awarded en route to Ph.D.)

Department of Energy Computational Science Graduate Fellow

Harvard University

2011 – 2015

Joint A.B. degree in Physics and Mathematics. Minor in Computer Science

Cum laude in field with High Honors in Physics and Mathematics

Selected Publications

† equal contribution, ‡ alphabetical authors. Full publication list available on [Google Scholar](https://scholar.google.com/citations?user=mirunaoprescu).

Preprints

- [1] Maresa Schröder, **Miruna Oprescu**, Stefan Feuerriegel, and Nathan Kallus. Causal inference on networks under misspecified exposure mappings: A partial identification framework. *arXiv preprint arXiv:2602.03459*, 2026.

Conference and Journal Publications

- [1] Ayush Khot[†], **Miruna Oprescu**[†], Maresa Schröder, Ai Kagawa, and Xihai Luo. Spatial deconfounder: Interference-aware deconfounding for spatial causal inference. In *International Conference on Machine Learning*. PMLR, 2026.
- [2] **Miruna Oprescu**, Brian Cho, and Nathan Kallus. Efficient adaptive experimentation with noncompliance. *Advances in Neural Information Processing Systems*, 38:42462–42498, 2026.
- [3] **Miruna Oprescu**, David Park, Xihai Luo, Shinjae Yoo, and Nathan Kallus. Gst-unet: A neural framework for spatiotemporal causal inference with time-varying confounding. *Advances in Neural Information Processing Systems*, 38:17295–17322, 2026.
- [4] **Miruna Oprescu** and Nathan Kallus. Estimating heterogeneous treatment effects by combining weak instruments and observational data. *Advances in Neural Information Processing Systems*, 37:118777–118806, 2025.

- [5] Andrew Bennett, Nathan Kallus, **Miruna Oprescu**[†], Wen Sun, and Kaiwen Wang[†]. Efficient and sharp off-policy evaluation in robust markov decision processes. *Advances in Neural Information Processing Systems*, 37:112962–113000, 2025.
- [6] Andrew Bennett[†], Nathan Kallus[†], and **Miruna Oprescu**[†]. Low-rank mdps with continuous action spaces. In *International Conference on Artificial Intelligence and Statistics*, pages 4069–4077. PMLR, 2024.
- [7] **Miruna Oprescu**, Jacob Dorn, Marah Ghoummaid, Andrew Jesson, Nathan Kallus, and Uri Shalit. B-learner: Quasi-oracle bounds on heterogeneous causal effects under hidden confounding. In *International Conference on Machine Learning*, pages 26599–26618. PMLR, 2023.
- [8] Nathan Kallus[†] and **Miruna Oprescu**[†]. Robust and agnostic learning of conditional distributional treatment effects. In *International Conference on Artificial Intelligence and Statistics*, pages 6037–6060. PMLR, 2023.
- [9] Soukayna Mouatadid, Paulo Orenstein, Genevieve Flaspohler, Judah Cohen, **Miruna Oprescu**, Ernest Fraenkel, and Lester Mackey. Adaptive bias correction for improved subseasonal forecasting. *Nature Communications*, 14(1):3482, 2023.
- [10] Keith Battocchi[‡], Eleanor Dillon[‡], Maggie Hei[‡], Greg Lewis[‡], **Miruna Oprescu**[‡], and Vasilis Syrgkanis[‡]. Estimating the long-term effects of novel treatments. *Advances in Neural Information Processing Systems*, 34:2925–2935, 2021.
- [11] **Miruna Oprescu**[†], Vasilis Syrgkanis[†], and Zhiwei Steven Wu[†]. Orthogonal random forest for causal inference. In *International Conference on Machine Learning*, pages 4932–4941. PMLR, 2019.
- [12] Vasilis Syrgkanis, Victor Lei, **Miruna Oprescu**, Maggie Hei, Keith Battocchi, and Greg Lewis. Machine learning estimation of heterogeneous treatment effects with instruments. *Advances in Neural Information Processing Systems*, 32:15193–15202, 2019. **Spotlight presentation.**
- [13] **Miruna Oprescu**[†], Vasilis Syrgkanis[†], Keith Battocchi[†], Maggie Hei[†], and Greg Lewis[†]. EconML: A Machine Learning Library for Estimating Heterogeneous Treatment Effects. In *CausalML Workshop, NeurIPS*, 2019. **Spotlight presentation.**
- [14] K Arbour, **M Oprescu**, J Hakim, H Rizvi, M Leiserson, M Ginsburg, A Plodkowski, J Sauter, I Preeshagul, S Gillett, et al. Multifactorial Model to Predict Response to PD-(L) 1 Blockade in Patients with High PD-L1 Metastatic Non-Small Cell Lung Cancer. *Journal of Thoracic Oncology*, 14(10):S290, 2019.

Research and Work Experience

Machine Learning Research Intern

Summers 2022, 2025

Netflix

- **2025, New York City:** Developed a structural causal model of subscription choice using quasi-experimental plan migration to estimate plan value, price sensitivity, and switching costs. Mentors: David Hubbard, Yonatan Gur.
- **2022, Los Gatos:** Developed causal ML methods combining weak instruments and observational data to estimate the effect of novel content on engagement [4]. Mentors: Sudeep Das, Aish Fenton.

Research Intern

Summer 2024

Brookhaven National Laboratory, Brookhaven, NY

- Developed GST-UNet [3], a neural framework for spatiotemporal causal inference under time-

varying confounding, integrating UNet architectures with iterative G-computation to estimate intervention effects in environmental and health applications. Mentor: Shinjae Yoo.

Senior Data and Applied Scientist

2017 – 2021

Microsoft Research, Cambridge, MA

- Conducted research in causal machine learning, developing methods for heterogeneous treatment effect estimation and policy evaluation, with results published at top venues [10, 11, 12, 13].
- Founded and maintained the open-source [EconML](#) library, implementing production-scale causal inference algorithms now widely used in industry and academia.
- Collaborated on projects in subseasonal weather forecasting and cancer modeling, applying ML techniques to real-world scientific data.

Software Development Engineer

2015 – 2017

Microsoft, Cambridge, MA

- Developed scalable, distributed machine learning algorithms as a core contributor to [MMLSpark](#), Microsoft's open-source ML library for Apache Spark.

Invited Talks

Causal Inference for Spatiotemporal Interventions 2025

SIAM Conference on Computational Science and Engineering

Reliable Treatment Effect Estimation Using Weak Instruments and Observational Data 2024

Workshop in Operations Research and Data Science (WORDS), Duke University

Uncertainty Quantification in Causal Inference: Sharp and Efficient Bounds on 2023

Heterogeneous Causal Effects Under Hidden Confounding

Brookhaven National Laboratory

Causal Inference and Machine Learning in Practice with EconML and CausalML 2021

SIGKDD

EconML: A Machine Learning Library for Estimating Heterogeneous Treatment Effects 2019

Open Data Science Conference East

Honors & Awards

Department of Energy Computational Science Graduate Fellowship 2022–2026

DOE Communicate Your Science & Engineering ([CYSE](#)) Award 2025

Meta Ph.D. Research Fellowship Finalist 2022

Cum laude, Harvard University; High Honors in Physics and Mathematics 2015

Derek C. Bok Award for Distinction in Teaching (Data Science), Harvard 2014

Teaching

Teaching Assistant, Cornell University 2021–2022

- Learning, Inference, and Decision Making from Data (Spring 2022)
- Applied Machine Learning (Fall 2021)

Teaching Fellow, Harvard University

2012–2014

Physics, Data Science, and Mathematics courses; recipient of the Derek C. Bok Award for Distinction in Teaching.

Professional Service

Director, Summer Science Program (SSP)

2023–Present

Reviewer: NeurIPS (Top Reviewer Award, 2025), ICML, AISTATS

2021–Present